

2018

COMPUTER SCIENCE

Paper : CSM-101

(Advances in Computer Architecture)

Full Marks : 70

*The figures in the margin indicate full marks.**Candidates are required to give their answers in their own words as far as practicable.*

Answer question nos. 1, 2 and any four questions from the rest.

1. Answer *any five* questions from the following :

2×5

- (a) Critically comment on the phrase— 'execution of instruction'.
- (b) Derive the formula for average access time in a three-level cache.
- (c) What are the desirable properties of RISC architecture?
- (d) What are different types of internal data forwarding? Give example.
- (e) What is the difference between a direct and indirect address instruction? How many references to memory are needed for each type of instruction to bring an operand into a processor register?
- (f) "Simultaneous transfer is supported by crossbar switch organization"— justify.
- (g) "Floating point addition has inherent parallel computing"— explain critically.
- (h) What do you mean by Future bus⁺ protocol?

2. Answer *any five* questions from the following :

- (a) Comment on the guidelines to convert any blocking multi-stage interconnection network to its equivalent non-blocking counterpart. 4
- (b) Estimate the effect of branch instructions on the performance of pipelined architectures with suitable set of parameters. 4
- (c) Comment on the characteristic features of any systolic array or processors. 4
- (d) What is the significance of virtual memory in advance computer architecture? Comment on the size of VM with proper justification. 4
- (e) Evaluate the maximum achievable speedup for a computing environment with large number of processors and programs having 10% sequential part. 4
- (f) Prove that a k-stage linear pipeline can be at most k times faster than that of a non-pipelined serial processor. Why buffer is used between consecutive two stages of a pipeline system? 3+1

Please Turn Over

- (g) For a computer system there is 8 KB of cache memory with 32 bytes per cache line. Computer has an address bus with 24 bits width and employing direct mapping type cache memory access. Now find the address format of this system. State the disadvantages of this mapping system. 3+1
- (h) Using a NUMA design, distinguish between clustered shared memory and global shared memory. 4
3. (a) Given that 'c' is the number of 0s in the initial collision vector, show that for any static pipeline and any modified state diagram—
the total number of states is no greater than 2^c .
Can you make similar statement for a dynamic pipeline? Justify.
- (b) What is the upper bound of minimal average latency of a static pipeline architecture? Justify your claim.
- (c) What is the maximum possible utilization of any stage of a pipeline that is computing initiations from some reservation table with latency cycle (2,3,7)? Justify. 3+3+4
4. (a) Compare and contrast SIMD array processor and systolic array processor.
- (b) Given with an algorithm of matrix multiplication on uni-processor system, modify the same to design the systolic architecture for matrix multiplication. Illustrate the design steps for matrices of size 4×4 . State the assumptions, if any. 4+6
5. (a) Write a parallel algorithm to add 32 numbers using a single stage 4-cube interconnection network when numbers are equally distributed over the processors. Clearly mention your assumptions.
- (b) What do you mean by hazard? What are different types of pipeline hazards? What are the alternative techniques to reduce the data hazard? 6+4
6. (a) What do you mean by "Parallelism by message passing" and 'illusions of message passing system'? Comment on the number and type of process states of any message passing system in a multiprocessor system.
- (b) What is multiport memory (MpM)? Briefly explain the effectiveness of MpM.
- (c) Comment on VLIW processor. 5+3+2
7. What are hazards in the pipeline based architecture? — Explain with a suitable example. Consider the following reservation table for a 4-stage pipeline with a clock cycle $t = 20\text{ ns}$ and calculate the following:

T_0	T_1	T_2	T_3	T_4	T_5	T_6	Stages
X					X		1
		X				X	2
	X		X				3
		X		X			4

- (a) What are the forbidden latencies and the initial collision vector?
- (b) Draw the state transition diagram for scheduling the pipeline.
- (c) Determine the MAL associated with the shortest greedy cycle.
- (d) Determine the pipeline throughput corresponding to the MAL and given t .
- (e) Determine the lower bound on the MAL for this pipeline.

2+2+2+2+1+1

(3)

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8. In the following block of computations, a and b are two external inputs and z is the final output. Two intermediate results are labelled x and y .

$$x \leftarrow a \times a; \quad y \leftarrow b \times b; \quad z \leftarrow \frac{x+y}{x-y};$$

- (a) Draw a dataflow graph for this code block.
- (b) Show a template implementation of the data flow graph in (a)
- (c) Indicate the events that can be done in parallel in the execution of the above code block.
- 4+3+3
9. (a) State the difference between vector processor and scalar processor. Explain vectorization process with respect to the following program fragment :
- ```
for(i=0;i<100;i++)
{
 A[i]=B[i]*C[i];
 D[i]=f(A[i]*X[i]);
 E[i]=D[i]/B[i]+A[i];
}
```
- (b) What are different routing schemes for Multistage interconnection networks? Explain with examples.

2+4+4